DATA MANAGEMENT DOCUMENT

DR NO COMIC ORDERING APP

How data is collected, used, stored and organized

A description of how customer and staff data moves through the mobile ordering app, based on the system architecture and the entity relationship diagram (ERD).

1 Introduction

1.1 Purpose

This document explains what data Dr No's Comic ordering app collects, why it is collected, how it is used, and where and how it is stored and organized. It covers both groups of users: customers who browse comics and place orders, and staff who manage the catalog and view customer orders.

1.2 How the app works in brief

The store is a simple ordering app. A customer creates an account with an email and password, browses the comics the store sells (the catalog), and places an order. There is no online payment. After the order is placed, a receipt is sent to the address the customer used to sign-in. Staff log in to add, update, delete and view catalog items, and to see the orders customers have placed

2 System Architecture

Figure 1 shows the app architecture. The app will be build using React Native with expo. The app shows different screens depending on the account's role. Every request goes over HTTPS to the backend API, which is the only part of the system that communicate with the database, file storage and email service

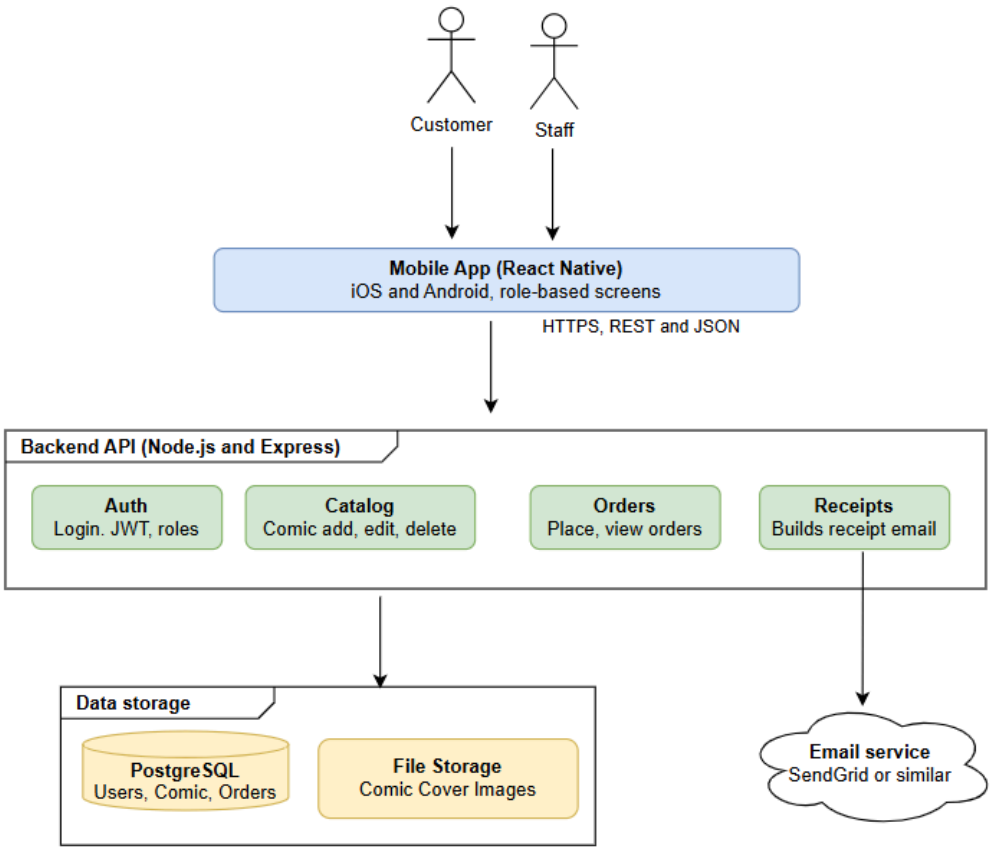


Figure 1. System Architecture

Component	**Role**	**Data it handles**
Mobile app (React Native with expo)	The screen the customer and staff use on iOS or Android. Shows customer screens or staff screens by role.	Form input (login details, order, comic details); temporary ordering list; login token kept on the device
Auth module	Registers accounts, checks login, issues a token (JWT) and enforces roles.	Email, password (stored only as a secure hash), role
Catalog module	Lets staff add, edit and delete comics and lets everyone view them.	Comic details and cover image links
Orders module	Places orders, saves order items, and lets customers and staff view orders.	Orders and order items
Receipts module	Builds the receipt for an order and sends it to the email service.	Order summary and the customer's email address
PostgreSQL	The main database. Holds all structured data.	Customer, Staff, Catalog, Order, Order_Item
File storage	Holds the comic cover image files.	The database holds the link.

Email service (SendGrid or similar)	External service that delivers receipt emails.	Recipient address and receipt content, only while sending

3 Data Model

Figure 2 shows the ERD. The structured data is organized into five tables, which are explained below.

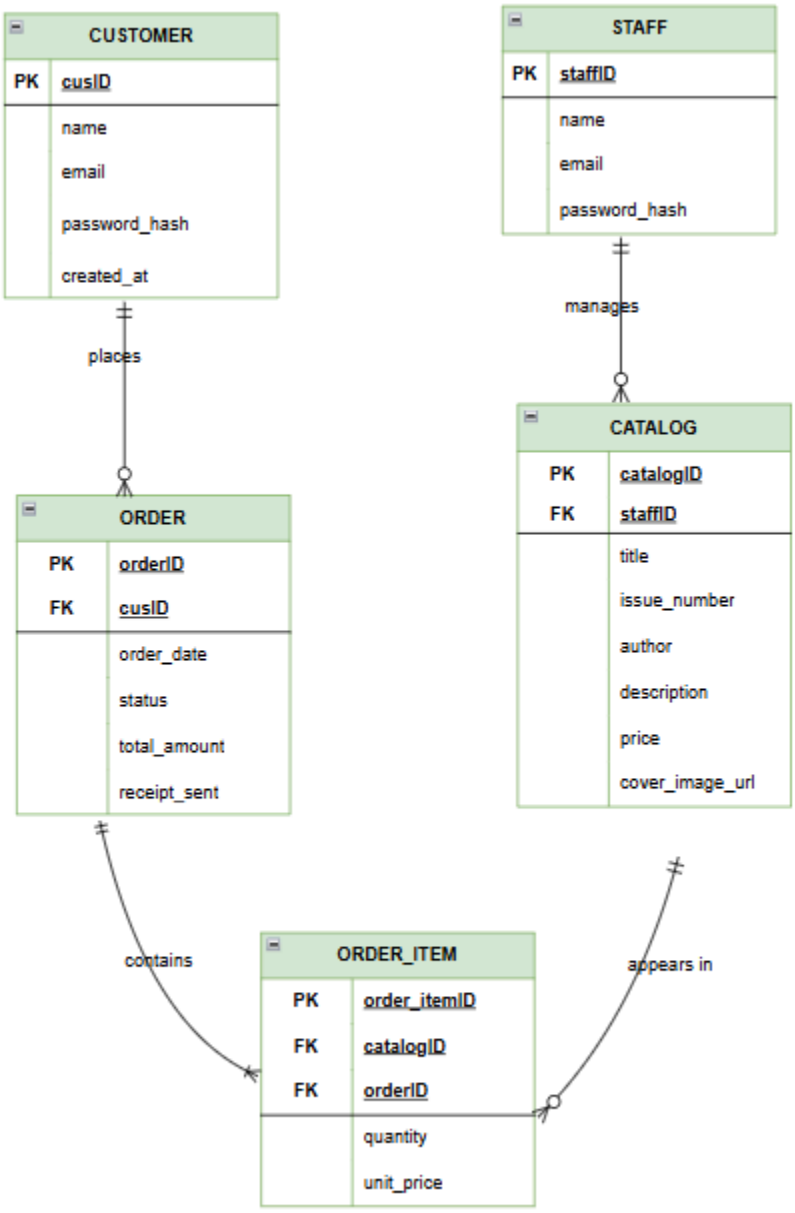


Figure 2: ERD

3.1 Entities

Entity	What it represent	Field	Maintained by
CUSTOMER	A person who has registered to browse and order comics.	cusID (PK), name, email, created_at, password_hash	The customer (created at sign-up)
STAFF	A store employee who manages comics and views orders.	staffID (PK), name, email, password_hash	Store owner or administrator (manually add by admin)
CATALOG	One comic the store sells.	catalogID (PK), staffID (FK), title, issue_number, author, description, price, cover_image_url	Staff
ORDER	One order placed by one customer.	orderID (PK), cusID (FK), order_date, status, total_amount, receipt_sent	The system, when the customer places an order
ORDER_ITEM	One line in an order: one comic and its quantity.	order_itemID (PK), catalogID (FK), orderID (FK), unit_price	The system, when the customer places an order

3.2 Relationships

- **Customer to Order (one to many).** One customer can place many orders. Each order belongs to exactly one customer, linked by cusID.
- **Order to Order_Item (one to many).** One order contains one or more order items. Each order item belongs to one order, linked by orderID.
- **Catalog to Order_Item (one to many).** One comic can appear in many order items across different orders. Each order item refers to one comic, linked by catalogID. Together with the previous relationship, Order_Item is the link that resolves the many-to-many relationship between orders and comics.
- **Staff to Catalog (one to many).** One staff member can manage many comics. Each comic records the staff member who added or last changed it, linked by staffID.

Staff are not linked to orders in the model. Staff can view all orders, but an order does not belong to a staff member.

4 How Data is Collected

4.1 Customer perspective

Data is collected from customers at two moments: when they sign up and when they place an order.

Data	Collected from	When	Stored in	Why It is needed
Name	Customer types it in	Account sign-up	CUSTOMER.name	To identify the customer on receipts and in the staff order view
Email address	Customer types it in	Account sign-up	CUSTOMER.email	It is the login name and the address the receipt is sent to
Password	Customer types it in	Sign-up and each login	Auth module, as a secure hash (never as plain text)	To check the customer's identity at login
Account creation time	Generated by the system	Account sign-up	CUSTOMER.created_at	Record keeping
Order list contents (comics and quantities)	Customer taps Add to order list	While shopping	Temporarily on the phone only	To build the order before it is placed
Order and items	Customer taps Place order; the system fills in the rest	At checkout	ORDER and ORDER_ITEM	To fulfil the order and keep order history
Order date, total, status	Generated by the system	At checkout	ORDER.order_date, total_amount, status	Receipt and order history
Price at time of purchase	Copied from CATALOG by the system	At checkout	ORDER_ITEM.unit_price	So old receipts stay correct if prices change later
Login token	Issued by the Auth module	After login	On the phone, until logout or expiry	So the customer stays signed in without retyping the password

4.2 Staff perspective

Staff accounts are not created through the public sign-up screen. The store owner or administrator creates them, so that customers can never register as staff. After that, data is collected from staff as they manage the catalog

Data	Collected from	When	Stored in	Why It is needed
Name and email	Administrator enters it	When the staff account is created	STAFF	To identify the staff member and let them log in
Password	Staff member	First setup and each login	Auth module, as a secure hash	To keep the staff area protected
Comic details (title, issue number, author, description, price, stock quantity)	Staff fill in the Add comic form	When adding or editing a comic	CATALOG	This is what customers browse and order
Cover image	Staff pick a photo from the phone gallery or camera	When adding or changing a comic	File storage, with the link in CATALOG.cover_image_url	Shown to customers next to the comic
Who changed a comic	Generated from the logged-in account	On every add or edit	CATALOG.staffID	Accountability for catalog changes
Deletion of a comic	Staff taps Delete	When removing a comic	Removes the CATALOG record	Keeps the catalog accurate

5 How Data Is Used

5.1 Customer Perspective

What the customer does	Data Used	What happens

Signs up	Name, email, password	The Auth module creates the account and stores the hashed password. A CUSTOMER row is saved.
Logs in	Email, password	The password is checked against the stored hash. If it matches, a token is returned and kept on the phone.
Browses comics	CATALOG rows, cover images	The app requests the list. The API returns comic details and cover image links, and the app loads each image from file storage.
Adds comics to order list	Comic and quantity	The order list is held on the phone. Nothing is saved on the server yet.
Places an order	Order list, customer ID from the token, current prices and stock	The API creates one ORDER and an ORDER_ITEM for each comic, copies the current price into unit_price, calculates total_amount
Receives a receipt	Order, items, customer name and email	The Receipts module builds the receipt and the email service delivers it to the customer's sign-in email. receipt_sent is then set to true.
Views past orders	ORDER and ORDER_ITEM rows for that customer only	The API filters by the customer ID in the token, so a customer can never see another customer's orders.

5.2 Staff Perspective

What the customer does	Data Used	What happens
Logs in	Email, password	The Auth module checks the credentials and confirms the account is a staff account. The token carries the staff role.
Adds a comic	Comic details and cover image	The image goes to file storage, and the API saves the comic in CATALOG with the image link and the staff member's ID.
Updates a comic	CATALOG row	Staff change details such as price or stock. The change shows to customers the next time they load the catalog.
Deletes a comic	CATALOG row	The comic is removed from the catalog. See section 8 for how this affects past orders.
Views the catalog	All CATALOG rows	Staff can list, search or filter by issue, author, price or availability.
Views customer orders	ORDER, ORDER_ITEM,	The API joins the tables and shows each order with the customer, comics, quantities, date and status.

	CUSTOMER name and email, comic titles	

6 Main Data Flows Step by Step

6.1 Customer places an order and receives a receipt

1. The customer opens the catalog. The app asks the API for the comics, and the API reads CATALOG from PostgreSQL.
2. The customer adds comics to the order list. The order list exists only on the phone.
3. The customer taps Place order. The app sends the order list and the login token to the Orders module over HTTPS.
4. The API checks the token
5. In a single database transaction, the API creates the ORDER row, creates one ORDER_ITEM row per comic with the current price, calculates total_amount. If any step fails, nothing is saved.
6. The Receipts module builds the receipt from the saved order and sends it, with the customer's email address, to the email service.
7. The email service delivers the receipt. The API sets receipt_sent to true on the order. The app shows an order confirmation

6.2 Staff adds a Comic

1. Staff log in. The token identifies them as staff, which unlocks the staff screens.
2. Staff fill in the comic form and pick a cover image from the phone.
3. The image is uploaded to file storage, which returns a link.
4. The app sends the comic details and the link to the Catalog module. The module checks the staff role and saves a new CATALOG row with the staff member's ID.
5. The comic is now visible to customers. The image file on the staff member's phone is no longer needed, because customers load it from file storage

6.3 Staff views customer orders

1. Staff open the Orders screen. The app sends the request with the staff token.
2. The API confirms the staff role, then reads ORDER, ORDER_ITEM, CUSTOMER and CATALOG together.
3. The app shows each order with the customer name and email, the comics ordered, quantities, total, date and status

7 How Data Is Stored and Organized

7.1 Storage locations

Location	What is stored	Notes
PostgreSQL database	The five tables: CUSTOMER, STAFF, CATALOG, ORDER and ORDER_ITEM	Main store of record. Related tables are connected by primary and foreign keys, which keeps the data consistent. Hosted on the server side, never on the phone.
File storage	Comic cover image files	Images are not placed in the database. The database keeps only the text link in cover_image_url, which keeps the database small and the images fast to load.
Auth module (inside the API)	Password hashes, and the logic for roles and tokens	Passwords are never stored as plain text. They are hashed with a one-way algorithm such as bcrypt.
Email service	Receipt email in transit	Used only to deliver the receipt. The store's own records of the order stay in PostgreSQL.
Customer's phone	Login token and temporary order list	No order history or personal records are kept on the phone. Logging out clears the token.

7.2 Organisation of the data

- **Every record has a unique ID.** cusID, staffID, catalogID, orderID and order_itemID identify each row and are used to connect tables.
- **Nothing is duplicated unnecessarily.** A customer's name is stored once in CUSTOMER and referenced from orders by cusID. The only deliberate copy is unit_price in ORDER_ITEM, which preserves the price paid at the time of purchase.
- **Order history is kept in the database.** Past orders are saved as ORDER and ORDER_ITEM rows, so customers can view them and staff can review them at any time.

7.3 Backup and retention

Data	**Proposed Handling**

Database	Automatic daily backups on the hosting plan, and a tested restore at least once before launch. Keep an export of the data owned by the client.
Customer accounts and orders	Kept while the account exists, so customers can see their order history. The client should decide how long to keep orders after an account is closed.
Cover images	Kept while the comic is in the catalog. Images of deleted comics may be removed from file storage.
Login tokens	Short-lived. They expire automatically and are removed from the phone at logout.

8 Access Control and Privacy

8.1 Who can see and change what

Data	Customer	Staff
Catalog (comics, prices, stock)	View only	View, add, edit, delete
Own account details	View	Not applicable
Own orders	View and place	View
Other customers' orders	No access	View
Customer names and emails	Own only	View, through the order screen
Staff accounts	No access	Own account only. Created by an administrator.
Passwords	Never shown to anyone	Never shown to anyone